CBF Python and Machine Learning Project:
Predicting Average House Prices in Madrid Neighbourhoods Using Socioeconomic Indicators

1. Project Objective

This project aims to develop and evaluate a machine learning model that uses 2025 socioeconomic indicators to predict average housing prices across Madrid neighbourhoods.

2. Dataset Description

The two raw datasets were used to obtain Madrid neighbourhood-level data. One dataset provided the socioeconomic indicators, while the other provided average housing prices. Following initial data inspection, the datasets were cleaned, relevant features were selected, and the data was merged. The resulting dataset contained 130 Madrid neighbourhoods (excluding the airport), 15 selected socioeconomic indicators, and the corresponding average housing price per square metre, all based on 2025 data.

As the original datasets were in Spanish and the data used for modelling was predominantly quantitative, Google Translate was primarily used to translate column names and relevant categorical information.

Table 1: Source information on the raw datasets used, which were published by Madrid City Council

Title	Description	Link
Panel of indicators of districts and neighborhoods of Madrid. Sociodemographic study (2020-2025)	This dataset provides a territorial overview of socioeconomic, health, demographic, educational, quality of life, housing, environmental, municipal facilities, citizen participation, and budget variables for the districts of Madrid. This information is also included at the city level and, where available, at the neighborhood level.	https://datos.madrid.es/dataset/300087-0-indicadores-distritos
Madrid City Council Database: Real Estate Registry Statistics: Annual Data by Districts. 4.2.1.6. Average Declared Housing Price (euros/m2) by District and Neighborhood according to Housing Type	N/A	https://servpub.madrid.es/CSEBD_WBINTER/seleccionSerie.html?numSerie=0504020100060

3. Tools and Libraries Used

<u>Python libraries:</u>

- Pandas: Data manipulation, cleaning and analysis
- NumPy: Numerical operations
- Matplotlib: Creating charts and data visualisations
- Seaborn: Creating statistical data visualisations
- Scikit-learn: Data preprocessing, building the machine learning model and evaluating model performance
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<u>Other tools:</u>

- Jupyter Notebook: To explore and clean the datasets, perform the analysis, and develop and evaluate the machine learning model
- Microsoft Excel: To support the initial analysis and selection of relevant socioeconomic indicator categories

4. Key Steps Followed

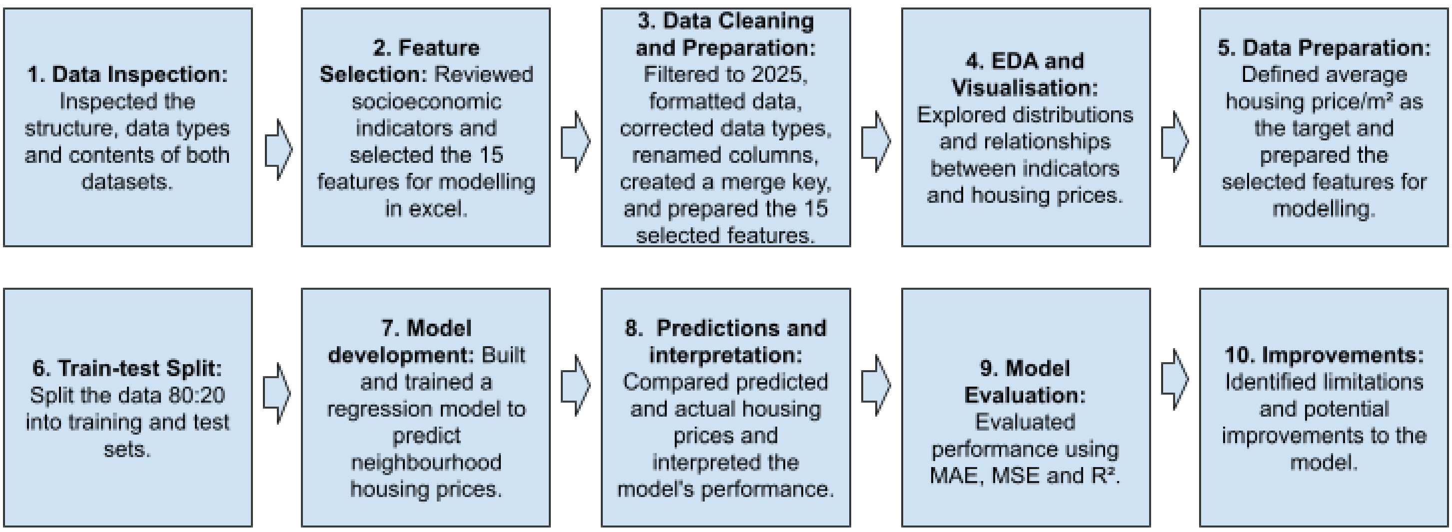


Fig 1: Flow diagram of stages undertaken

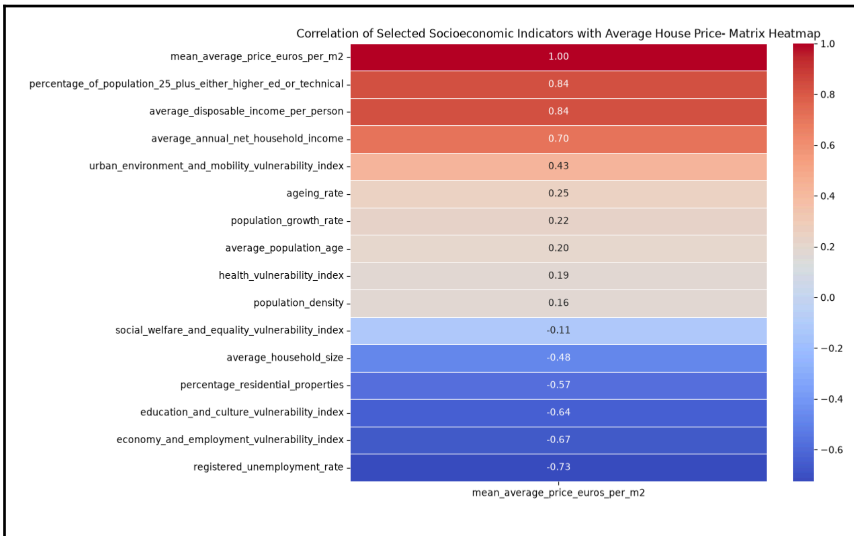
5. Results

A Linear Regression model was selected as the target variable is numerical. The dataset was split 80:20 into training and test data, a random state used for reproducibility. Approximately 30% of prediction values were within 5% of actual housing prices and 77% were within 20%. While the model performed well for many neighbourhoods, several predictions showed larger errors, suggesting scope for improvement.

The MAE and R^2 score indicate reasonably accurate predictions and that the model explains a substantial proportion of the variation in housing prices. The difference between MAE and RMSE, alongside the MSE, indicates that a smaller number of neighbourhoods had larger prediction errors.

Table 2: Model Performance Metrics

Metric	Value
Mean	€5,260/m²
Mean Absolute Error (MAE)	€713/ m²
Mean Squared Error (MSE)	897,950(€/m²)²
Root Mean Squared Error (RMSE)	€948 /m²
R^2 Score	0.85



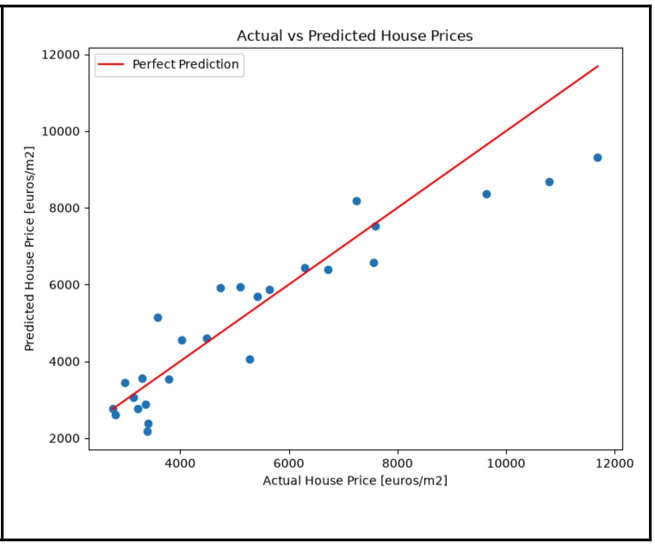


Fig 2: 'Heatmap of Correlation between Average House Price & the selected 15 Socioeconomic Indicators', which displayed a range of correlation strengths. In future iterations, feature selection could be refined by focusing on the indicators with the strongest relationships

Fig 3: Actual vs Predicted Average Housing Prices, showing the model's predicted values compared with the actual values for the test dataset.

6. Challenges Encountered

The raw datasets required substantial cleaning and reformatting before modelling. Working with raw data was a deliberate challenge and required time to explore and determine appropriate preparation methods. Initial challenges included CSV metadata, the unexpected semicolon separator, and Spanish language data. Feature selection was challenging due to the number of available indicators and limited time to conduct a literature review. Finally, as multiple features were used, the regression could not be represented in a simple 2D plot, therefore an actual-versus-predicted plot was used to visualise model performance.

7. Suggestions for Improvement

The model is limited by its 130 observations from a single year, feature selection without a supporting literature review, and its ability to identify relationships rather than causation.

Future improvements that may reduce prediction errors and improve the model's predictive performance could include:

- Outlier treatment: Investigate high-value outliers and their influence on model performance.
- Alternative models: Compare Linear Regression with other regression models capable of capturing more complex or non-linear relationships.
- Feature scaling: Standardise features measured on different scales where appropriate.
- Feature engineering: Create additional derived features that may better represent relationships within the data.
- Feature selection: Refine the 15 selected features using further research and analysis, including removing consistently weak predictors.
- Historical data: Incorporate previous years to capture longer-term trends and potentially improve future predictions.

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